16

No of additional sheets 3

LOURDES MATHA COLLEGE OF SCIENCE & TECHNOLOGY

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ANSWER BOOK

Name Adwaith. A. M Class No. 05 Branch MCA

Subject Data science and machine learning Semester S3

Mark Tabulation

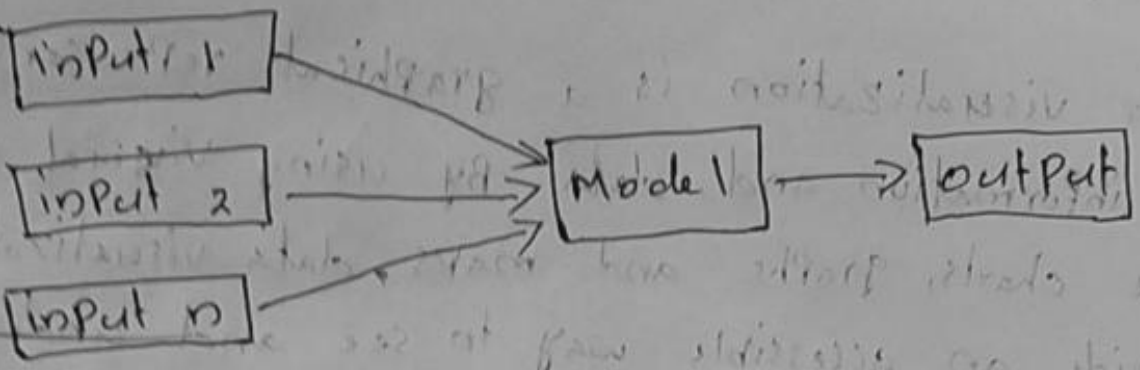
Mark Divisions	Q.1	Q.2	Q.3	Q.4	Q.5	Q.6	Q.7	Q.8	Q.9	Q.10	Q.11	Q.12	Q.13	Q.14	Q.15
	2½	3		2		3		1	4½						
Total															
Grand Total													16		

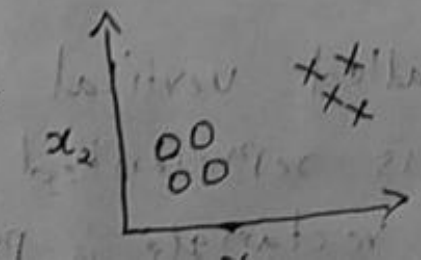
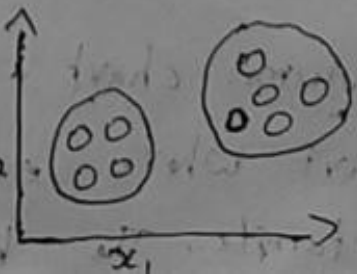
1) Data science is the collection of techniques used to extract value from data. It has become an essential tool for any organisation that collects, stores and processes data as part of its operations. Data science techniques find useful patterns connections and relationships within data. Data science is also commonly refered to as knowledge discovery, machine learning, predictive analytics and data mining. However each of these terms have a different meaning depending on the context. Data science enables companies to efficiently understand gigantic data from multiple sources and derive valuable

insights to make smarter data driven decisions. Data science is widely used in various industry domains including marketing, healthcare, finance, banking, Policy work etc. The principal purpose of data science is to find Pattern within data. It uses various statistical techniques to analyse and draw insights from the data.

~~from data extraction, cleaning and processing~~

Q, Machine learning is a type of artificial intelligence that allows software applications to become more accurate at Predicting outcomes without being explicitly Programmed to do so. Machine learning algorithms use historical data as input to predict new output values. Recommendation engines are a common use case for machine learning. other popular uses include fraud detection, spam filtering, malware threat detection, business process automation and Predictive maintenance. Machine learning is important because it gives enterprises a view of trends in customer behaviour and business operational patterns as well as supports the development of new products. Many of today's leading companies such as facebook, google make machine learning a central part of their operations. Machine learning has become a significant competitive differentiator for many companies.



6, supervised learning	unsupervised learning
* input data is labelled.	* input data is unlabelled
* Has a feedback mechanism	* Has no feedback mechanism
* Data is classified based on the training dataset	* Assigns properties of given data to classify it
* Divided into regression and classification	* Divided into clustering and association.
* used for prediction	* used for analysis
* Algorithms include : decision trees, logistic regressions, support vector machine	* Algorithms include : k-means clustering, Hierarchical clustering, APriori algorithm.
* A known no: of classes	* A unknown no: of classes
* 	* 

9, Data visualization is a graphical representation of information and data. By using visual elements like charts, graphs and maps, data visualization tools provide an accessible way to see and understand trends, outliers and patterns in data.

~~Benefits~~

~~* Powerful way to explore data with presentable results.~~

~~* Primary use is the pre processing portion of the~~

Data visualization techniques

* Box Plots
* Histograms
* Heat maps
* Charts
* Tree maps
* Word cloud

Methods used to visualize data

1, Column chart: It is also called vertical bar chart where each category is represented by a rectangle. The height of the rectangle is proportional

to the values that are plotted.

2, Bar graph :- It has rectangular bars in which the lengths are proportional to the values which are represented. diagrams?

3, stacked bar graph :- It is a bar style graph that has various components stacked together so that apart from the bar, the components can also be compared to each other

4, stacked column chart :- It is similar to the stacked bar, however the data is stacked horizontally

5, Area chart :- It combines the line chart and bar chart to show how the numeric values of one or more groups change over the progress of a viable data.

6, Dual axis chart :- It combines a column chart and a line chart and then compares the two variables.

7, Line graph :- The data points are connected thro a straight line, therefore creating a representation the changing trend.

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8, Mekko chart: It can be called two dimens stacked chart with varying column widths

9, Pie chart: It is a chart where various compone of a dataset are presented in the form of a which represents their proportion in the entir dataset.

10, waterfall chart: with the help of this chart, t increasing effect of sequentially introduced positive or negative values can be understood.

11, Bubble chart: It is a multi variable graph tha is hybrid of scatter plot and a proportional area chart

12, scatter plot chart:

13, Bullet graph

14, Funnel chart

15, Heat map.

8, Data science process :

it is the method of discovering useful relationships and patterns in data.

steps

* Prior knowledge
* Data preparation
* Modelling
* Application

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